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CHARACTERISTICS OF HUMAN INFLUENZA AH1N1, AH3N2, AND B VIRUSES ISOLATED FEBRUARY TO AUGUST 2008

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Summary of antigenic and genetic characteristics of influenza A H1N1, H3N2 and B viruses isolated February to July 2008

Approximately 600 influenza viruses, isolated in 35 countries between February and July 2008, were characterized. Approximately half were A H1N1 (53%), and 10% were H3N2. Influenza B viruses accounted for 37%, of which the majority (93%) were of the B/Yamagata lineage (Table 1).

H1N1 viruses

The majority of H1N1 viruses were antigenically closely related to the MDCK cell-grown reference virus A/Netherlands/345/2007 or the vaccine virus A/Brisbane/59/2007 (Table 2 A and B). Differences in reactivity with antisera to these reference viruses correlated well with a change at position 186 in HA, D186N/V, which was present in most egg isolates. A small proportion gave relatively low HI titres with the various reference antisera, in particular recent isolates from Hong Kong. Very few were A/New Caledonia/20/99-like.

The majority of **HA and NA sequences** fell within clade 2B, represented by A/Brisbane/59/2007 (Figures 1 and 2; Table 3), and, in general, there has been relatively little change in the HA or NA sequences during 2008. The HAs of a number of recent isolates from southern hemisphere countries comprised a subgroup characterized by the changes N183S, G185S/N and A189T. A number of sequences, especially of viruses isolated in Japan and recently in Hong Kong fell within clade 2C, represented by A/Hong Kong/2652/2006. All the clade 2 HA sequences possessed the K140E substitution, typical of A/Solomon Islands/3/2006-like and A/Brisbane/59/2007-like viruses.

Drug resistance

Overall, 30% of the H1N1 isolates were shown to possess high level resistance to oseltamivir, either by enzyme inhibition assay or the presence of the H275Y mutation in the NA (H274Y in N2 numbering), or both (Table 1). Data collected by the EISS/VIRGIL network for the season as a whole in Europe gave an average of approximately 24% resistance, ranging from no detections in some countries to 68% in Norway, and there was an increasing trend to April/May with an average of approximately 56% by the end of the season. Oseltamivir-resistant viruses detected among African isolates from February onwards increased to become predominant in a number of African countries, including South Africa where they accounted for 100% of viruses isolated during May to August.

The majority of oseltamivir-resistant N1 sequences fell within the “European” cluster within clade 2B, characterised by the change D354G in addition to H275Y (Figure 2). Only a few of the clade 2C NAs were shown to be resistant to oseltamivir.

The majority of oseltamivir-resistant (and -sensitive) viruses tested were shown to retain sensitivity to zanamivir and amantadine/rimantadine, with the exception of viruses within clade 2C, represented by A/HK/2652/06, which possessed the resistance mutation S31N in M2 and a few were resistant to both oseltamivir and amantadine/rimantadine.

Oseltamivir-resistant and -sensitive viruses were antigenically indistinguishable.

H3N2 viruses

In **HI tests**, although some of the H3N2 viruses were shown to be antigenically closely related to A/Brisbane/10/2007, most gave low HI titres, relative to the homologous virus, with antisera to this egg-grown virus and the vaccine strain A/Uruguay/716/2007, as well as to other recent reference viruses (Table 4 A and B). In general, they gave patterns of HI reactivity similar to those of the MDCK cell-grown reference viruses, A/Trieste/25c/2007, A/Finland/9/2008, A/Hong Kong/1050/2008 and/or A/Johannesburg/15/2008 (which also gave relatively low homologous HI titres with their corresponding ferret antisera, compared with egg-passaged reference viruses).

Neutralization tests, which did not distinguish between egg-grown (e.g. A/Brisbane/10/2007) and MDCK cell-grown (e.g. A/Trieste/25c/2007) viruses, showed that the small number of recent isolates tested were antigenically closely related to A/Brisbane/10/2007 and A/Uruguay/716/2007 (Table 5).

The **HA sequences** of recent isolates fell within the clade represented by A/Brisbane/10/2007 and characterized by the amino acid changes G50E and K140I, relative to A/Wisconsin/67/2005. Some increased heterogeneity, particularly with respect to residue 173, distinguished three principal subgroups, represented by A/Johannesburg/5/2008 (K173Q), A/Johannesburg/15/2008 (K173N plus L3F, K83N, L157S,) and, of lesser prominence, A/Wisconsin/3/2007 (K173E), (Figure 3).

The **NA sequences** of these viruses showed corresponding drift – e.g. those represented by A/Johannesburg/15/2008 possessed additional changes D147N and P45S (Figure 4).

Amantadine resistance. M2 sequences of selected viruses showed that, as for earlier A/Brisbane/10/2007-like isolates, they possessed the amantadine resistance mutation S31N in M2.

B viruses

The majority (93%) of B viruses received were of the B/Yamagata lineage, although higher proportions of B/Victoria-lineage viruses have been reported in some countries.

In **HI tests**, many of the **B/Yamagata-lineage viruses** were shown to be antigenically closely related to the vaccine strains, B/Florida/4/2006 and B/Brisbane/3/2007 (and B/Sendai-H/114/2007), whereas others were more closely related to the recent reference viruses, B/England/145/2008(egg), B/Barcelona/143/2008(MDCK), B/Valladolid/18/2008(MDCK) and/or B/Bangladesh/3333/2007(egg) (Table 6A and B).

HA sequences of recent B/Yamagata-lineage isolates were similar to those of viruses isolated during 2007 and fell into 3 principal clades, two of which are represented by the vaccine viruses B/Florida/4/2006 and B/Brisbane/3/2007 (and B/Sendai-H/114/2007), respectively, and the third by B/England/145/2008 and B/Bangladesh/3333/2007 (Figure 5). The latter accounted for about half of the sequences determined for recent isolates. The three clades were antigenically indistinguishable

NA sequences fell into corresponding subgroups within two clades; those represented by the vaccine viruses were similar and distinguished from those within the B/England/145/2008 clade by 7 amino acid differences (Figure 6; Table 8).

Of the few **B/Victoria-lineage viruses**, most were antigenically closely related to the MDCK cell-grown reference virus B/Hong Kong/45/2005, but gave reduced HI titres with antiserum to the egg-grown vaccine strain B/Malaysia/2506/2004 and more recent reference virus B/Victoria/304/2006, as previously observed (Table 7). Some recent egg-grown viruses were shown to be similar to the latter viruses.

Their **HA sequences** were close to those of B/Victoria-lineage viruses isolated during 2007 and of B/Malaysia/2506/2004 (and B/HK/45/2005), close to the consensus sequence (Figure 7; Table 8).

NA sequences were somewhat more diverse; most were similar to those obtained for the majority of viruses isolated during 2006 and 2007, and differed from that of B/Malaysia/2506/2004 by 8 amino acids; some others e.g. of B/Trieste/8/2008 were closer to that of B/Malaysia/2506/2004, with additional changes D342G and M375K (Figure 6; Table 8).

Table 1. Human influenza isolates, February 2008 - present

Country	H1N1		H3N2	B	
	A/NC/20/99 ¹	A/Bris/59/07		Yamagata lineage	Victoria lineage
February					
Belarus	3	4			
Belgium		5 2			
Cameroon		3 1			
Estonia		2 1			
France		3	4	17	1
Georgia		4 1			
Germany		1	1	1	
Greece		6		1	
Iceland		1		8	
Italy		6 1		9	2
Madagascar		27		1	
Morocco		1		1	
Poland		1		4	
Portugal				1	
Romania		11		11	
Russia		13 4	1	3	
Serbia and Montenegro		3		1	
Slovakia		4			
Spain				14	
Switzerland		1 1	1	3	
Ukraine	1	48 15		3	1
March					
Belgium		17 9			
Cameroon		2 2			
Finland		2			
France		2	2	12	1
Georgia		2			
Germany		16 9	3	16	1
Hong Kong		3	2	1	2
Iceland		5	4	4	
Ireland				1	
Latvia				3	
Madagascar		7			
Morocco				1	
Norway		4 2			
Poland		2 1		2	
Russia		8 2		5	
Slovakia				1	
Spain				1	
Switzerland		1 1	1	4	2
Ukraine		8 2		5	1
April					
Argentina		1		2	
Denmark			5	3	
Finland				1	
France				2	
Hong Kong			1	1	1
Iceland			13	2	
Ireland				1	
Latvia				3	
Madagascar		3			
Morocco					1
Norway		5 5	4	7	
Russia		7 3	5	4	
Seychelles		3			
Slovakia				2	
South Africa			1		1
Spain				3	
Sweden			3		
United Kingdom				2	
May					
Argentina		1 1		1	
Cameroon		2 2			
Finland				1	
France			1	3	
Ghana		10 2			
Hong Kong				1	
Iceland			2	1	
Madagascar		2		1	
Norway				6	
South Africa		4 4		2	
Sweden				1	
June					
Argentina		3 1		6	
France		1	1	1	
Ghana		3 2			
Madagascar		2			
Mauritius				1	
Norway				3	
Seychelles		5 5			
South Africa		9 7	2		1
July					
Ghana		9 7	1		
Hong Kong		6	2	2	
Mauritius		1	1	1	
South Africa		13 13		4	
Total = 595	4	313 93	61	202	15
		317 (53%)	(10%)	217 (37%)	

oseltamivir resistant; 1. A/NC/20/99 includes viruses that are A/New Caledonia/20/99-like and have HAs that may fall within clades 1 or 2

Table 2A. Antigenic analyses of influenza A H1N1 viruses

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹								
			Post infecton ferret sera								
			A/NC	A/Thess	² A/HK	² A/SI	² A/Fuk	² A/Neth	² A/Bris	² A/Egypt	² A/St. Pet
			20/99	24/05	2652/06	3/06	141/06	345/07	59/07	10/07	5/08
			F18/06	F10/05	F14/06	F8/07	NIB 1/07	F23/07	F2/08	F4/08	F13/08
REFERENCE VIRUSES											
A/New Caledonia/20/99	09/06/99	Ex	320	320	<	40	40	80	80	40	80
A/Thessaloniki/24/05	03/02/05	E2 \3	320	2560	80	80	80	160	160	80	80
A/Hong Kong/2652/2006	17/07/06	E2 \2	<	<	320	320	320	80	640	320	320
A/Solomon Islands/3/2006	21/08/06	E3 \2	80	160	320	640	640	640	1280	320	640
A/Fukushima/141/2006	07/02/06	E4 \2	40	80	160	640	640	160	2560	640	640
A/Netherlands/345/2007	12/10/07	MDCK2 \4	40	160	160	320	160	1280	640	320	320
A/Brisbane/59/2007	01/07/07	E2 \1	<	40	80	320	320	160	1280	640	640
A/Egypt/10/2007	30/10/07	E1 \2	<	40	80	160	320	80	1280	640	640
A/St. Petersburg/5/2008	13/02/08	E2 \2	80	80	640	2560	320	640	1280	1280	640
TEST VIRUSES											
A/Kiev/245/2008	06/02/08	MDCK3 \1	40	80	80	160	80	640	320	160	ND
A/Cameroon/021/2008	07/02/08	MDCKx \1	40	40	40	<	ND	640	80	160	160
A/Cameroon/023/2008	12/02/08	MDCKx \1	80	160	160	160	ND	1280	80	160	320
A/Cameroon/025/2008	19/02/08	MDCKx \1	40	40	160	160	ND	160	160	320	320
A/Cameroon/044/2008	26/03/08	MDCKx \1	80	80	160	320	ND	1280	320	320	320
A/Cameroon/073/2008	16/05/08	MDCKx \1	40	40	40	40	ND	640	80	160	160
A/Cameroon/080/2008	31/05/08	MDCKx \1	80	160	160	320	ND	1280	160	160	320
A/Georgia/727/2008	12/02/08	MDCK2 \1	40	40	80	160	ND	160	160	320	ND
A/Minsk/179/2008	20/02/08	MDCK1 \1	<	<	40	40	<	640	320	320	160
A/Berlin/47/2008	03/03/08	MDCKx \1	<	160	160	320	160	640	320	160	ND
A/Ukraine/372/2008	03/03/08	MDCK1 \1	80	80	160	320	320	640	160	160	320
A/Norway/644/2008	03/03/08	MDCK \1	<	<	40	80	ND	160	80	160	ND
A/Hong Kong/555/2008	06/03/08	MDCK2 \1	80	80	160	40	640	320	320	320	320
A/Bialystok-PL/1/2008	10/03/08	MDCK1 \1	40	<	80	160	320	160	640	640	320
A/Geneva/3038/2008	10/03/08	MDCK1 \1	<	160	160	160	80	640	320	160	ND
A/Norway/815/2008	12/03/08	MDCK \1	80	160	160	160	ND	640	80	160	ND
A/Kursk/1/2008	21/04/08	C2 \1	<	40	80	160	160	160	640	320	ND
A/Seychelles/2010/2008	23/04/08	MDCK1 \1	80	80	80	160	ND	640	160	160	ND
A/Ghana/122/2008	May-08	MDCK \1	40	80	160	160	640	160	160	160	160

Vaccine strain

¹ < = <40; ² 140E in HA1, oseltamivir resistant; ND = Not done.

Table 2B. Antigenic analyses of influenza A H1N1 viruses

Haemagglutination inhibition titre ¹											
Viruses	Collection Date	Passage History	Post infection ferret sera								
			A/NC	A/Thess	² A/HK	² A/SI	² A/Neth	² A/Bris	² A/Egypt	² A/SP	² A/SP
			20/99	24/05	2652/06	3/06	345/07	59/07	10/07	5/08	12/08
			F22/06	F10/05	F14/06	F8/07	F23/07	F2/08	F4/08	F13/08	F17/08
REFERENCE VIRUSES											
A/New Caledonia/20/99	09/06/99	Ex	640	640	40	80	80	40	80	80	40
A/Thessaloniki/24/05	03/02/05	E2 V3	1280	2560	80	40	160	40	160	80	80
A/Hong Kong/2652/2006	17/7/06	E2 V2	40	<	320	320	80	320	320	320	320
A/Solomon Islands/3/2006	21/8/06	E3 V2	160	160	320	640	640	320	160	320	160
A/Netherlands/345/2007	12/10/07	MDCK2 V4	160	160	160	320	1280	160	320	320	320
A/Brisbane/59/2007	1/7/07	E2 V1	40	40	160	320	160	640	640	640	320
A/Egypt/10/2007	30/10/07	E1 V2	40	40	160	320	160	320	640	320	320
A/St. Petersburg/5/2008	13/2/08	E2 V2	80	80	320	1280	640	640	640	640	640
A/St. Petersburg/12/2008	13/2/08	E2 V2	40	40	160	320	160	320	640	320	320
TEST VIRUSES											
A/Seychelles/2239/08	03/06/08	MDCK2 V1	40	80	160	160	80	160	160	320	ND
A/Lyon/1059/08	09/06/08	MDCK2 V1	80	160	160	160	640	160	320	320	ND
A/Johannesburg/36/2008	10/06/08	MDCKx V1	40	<	40	40	640	160	160	160	ND
A/Chubut/102197/2008	11/06/08	M1 V1	80	160	160	320	160	160	320	320	ND
A/Madagascar/2360/08	16/06/08	MDCK1 V1	160	160	320	320	160	320	320	320	ND
A/Ghana/152/2008	17/6/2008	MDCKx V1	80	80	320	160	640	160	160	320	320
A/La Plata/535993/2008	18/06/08	M1 V1	80	160	160	160	640	160	160	160	ND
A/Ghana/163/2008	20/6/08	MDCKx V1	160	160	320	320	1280	160	320	320	320
A/Hong Kong/1404/2008	07/07/08	MDCK2 V1	40	40	80	80	320	40	80	ND	40
A/Hong Kong/1496/2008	14/07/08	MDCK2 V1	<	<	80	80	80	80	160	ND	80
A/Hong Kong/1776/2008	23/07/08	MDCK2 V1	<	<	80	40	160	40	80	ND	40
A/Mauritius/403/2008	09/07/08	MDCK1 V1	<	<	40	<	320	80	160	160	160
A/Ghana/02/2008	25/7/08	MDCKx V1	<	<	80	80	160	80	160	160	160
A/Ghana/09/2008	25/7/08	MDCKx V1	<	<	80	80	160	160	160	160	160
A/Cape Town/119/2008	July-08	MDCK V1	80	80	160	160	640	160	320	ND	320
A/Cape Town/124/2008	July-08	MDCK V1	80	160	160	160	1280	160	160	ND	320
A/Johannesburg/165/2008	July-08	MDCK V1	40	40	80	80	640	80	160	ND	160
A/KwaZulu Natal/272/2008	July-08	MDCK V1	40	80	160	160	640	80	160	ND	160
A/Johannesburg/340/2008	July-08	MDCK V1	80	80	160	160	640	160	160	ND	320
A/Osaka/8/2008	18/01/08	MDCK1+2 V1	<	<	80	40	80	40	80	ND	80
A/Shiga/8/2008	30/01/08	E3+2 V1E	<	<	160	160	40	80	320	ND	320
A/Shiga/8/2008	30/01/08	MDCK1+2 V1	40	40	160	80	80	40	160	ND	160
A/Perth/200/2008	22/02/08	E3 V1E	<	<	640	320	160	320	320	ND	320
A/Sapporo/33/2008	03/04/08	MDCK1+2 V1	40	40	160	320	160	80	320	ND	160
A/Santiago/5593/2008	17/04/08	E1/E2 V1E	<	<	80	160	40	160	320	ND	320
A/Brazil/1059/2008	06/06/08	C2/C2 V1	40	40	80	160	160	320	640	ND	320
A/Victoria/501/2008	10/07/08	E2 V1E	<	<	80	80	160	40	160	ND	80

Vaccine strain

¹ < = <40; ² 140E in HA1, oseltamivir resistant; ND = Not done.

Figure 1. Phylogenetic comparison of H1 HA genes

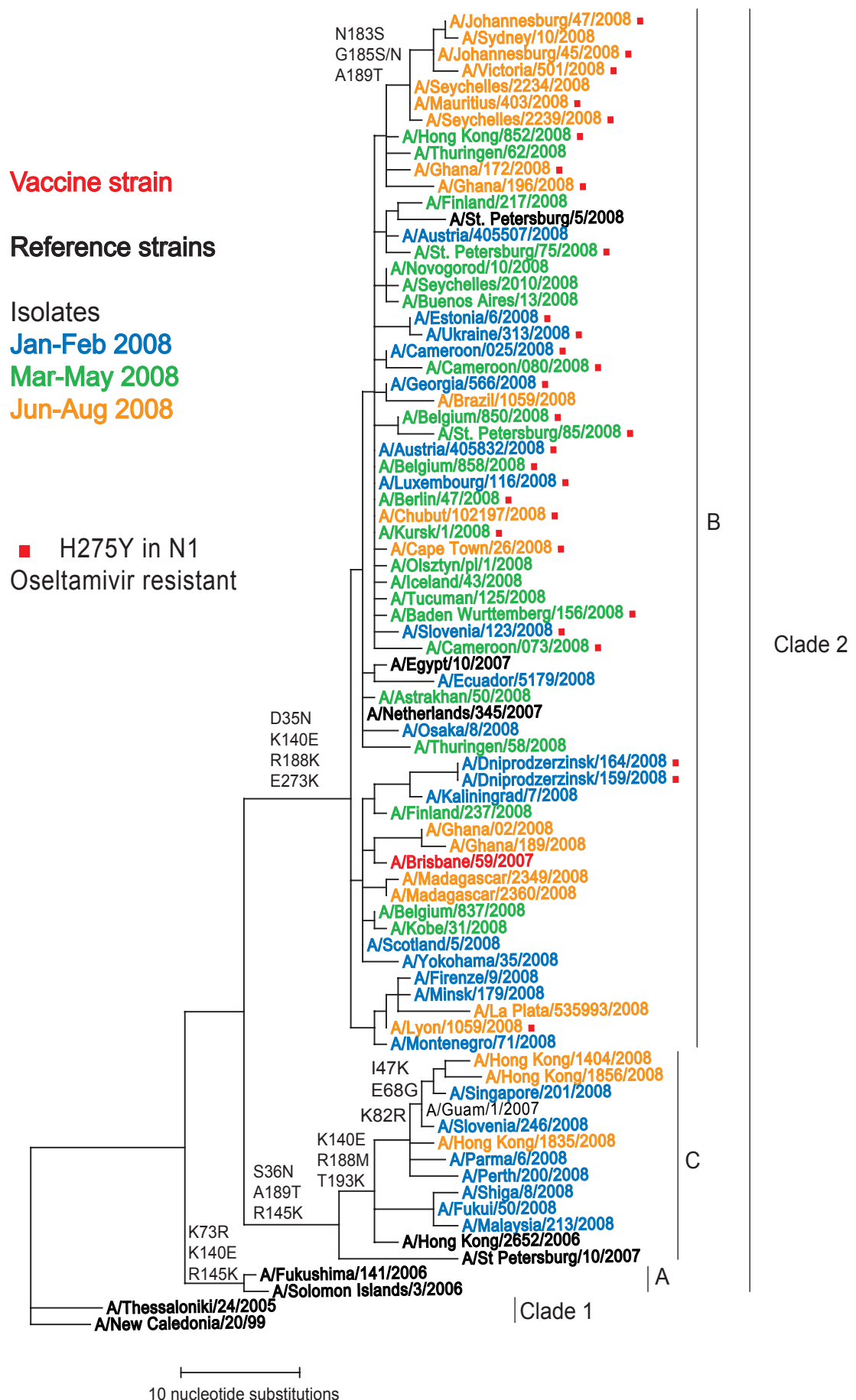


Figure 2. Phylogenetic comparison of N1 neurmindase genes

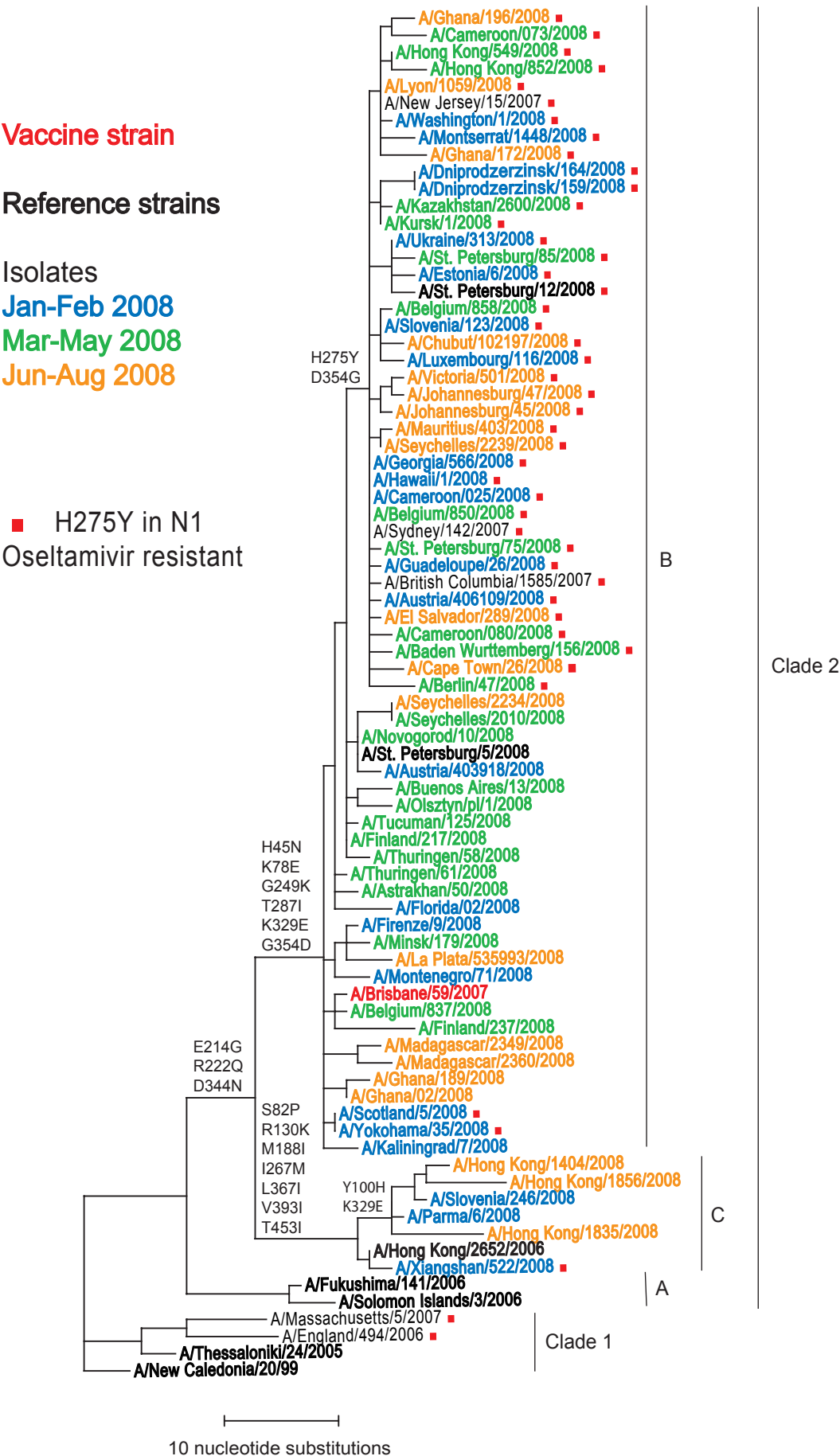


Table 3. Amino acid changes characteristic of the recent principal H1N1 sequence variants

Clade	Representative strain	Amino acid changes	
		HA	NA
2B	A/Brisbane/59/07	D35N	H45N
		K145R	K78E
		R188K	G249K
		E273K	T287I
		(N183S)	K329E
		(G185S/N/A)	(H275Y)
		(A189T)	(G354D)
2C	A/Hong Kong/2652/06	S36N	S82P
		R188M	R130K
		A189T	M188I
		T193K	I267M
		(N244S)	L367I
		(I47K)	V393I
		(E68G)	T453I
		(K82R)	(Y100H) (K329E)

Table 4A. Antigenic analyses of influenza A H3N2 viruses

Haemagglutination inhibition titre ¹												
Viruses	Collection Date	Passage History	Post infection ferret sera									
			A/Wis	A/Nepal	A/Trieste	A/Prag	A/Tri	A/Wis	A/Hen	A/Bris	A/Uru	A/Fin
			67/05 F6/06	921/06 F2/07	25c/07 F5/07	3/07 F9/07	25E/07 F11/07	3/07 F15/07	147/07 F16/07	10/07 F18/07	716/07 F3/07	9/08 F12/08
REFERENCE VIRUSES												
A/Wisconsin/67/2005	31/8/05	SpfCk3, E3 \3	5120	5120	2560	640	2560	1280	5120	2560	5120	2560
A/Nepal/921/2006	11/7/06	E2 \2	5120	5120	5120	1280	2560	1280	5120	1280	5120	5120
A/Trieste/25c/2007	Jan-07	MDCKx \2	80	80	80	<	160	160	80	160	160	80
A/Prague/3/2007	Feb-07	MDCK3, E4 \2	2560	5120	2560	1280	2560	640	5120	640	2560	2560
A/Trieste/25E/2007	Jan-07	MDCKx \2,E1	2560	2560	2560	640	1280	640	5120	1280	2560	2560
A/Wisconsin/3/2007	21/1/07	Ex \2	5120	5120	5120	1280	2560	2560	5120	2560	5120	2560
A/Henan/Jinshui/147/2007	unknown	E3 \1	2560	2560	2560	640	1280	640	5120	1280	2560	2560
A/Brisbane/10/2007	6/2/07	E2 \3	2560	2560	2560	640	2560	640	5120	1280	2560	2560
A/Uruguay/716/2007	21/6/07	SpfCk1, E3 \1	2560	2560	2560	640	2560	640	5120	1280	2560	2560
A/Finland/9/2008	7/1/08	MDCK2 \1	40	80	160	40	160	320	160	320	160	320
TEST VIRUSES												
A/Paris/2030/2008	Feb-08	MDCKx \1	80	80	80	<	160	160	80	320	160	80
A/Lyon/669/08	14/02/08	MDCK2 \1	40	80	80	40	80	80	80	80	80	80
A/Netherlands/177/2008	18/02/08	xMDCK2 \1	160	80	320	80	160	320	320	640	320	640
A/Berlin/50/2008	11/03/08	MDCKx \1	80	80	40	<	80	80	80	160	80	ND
A/Hong Kong/796/2008	15/03/08	MDCK2 \1	320	160	160	40	320	320	640	640	160	320
A/Khabarovsk/5/2008	14/04/08	C6 \1	160	80	80	<	80	160	160	160	160	80
A/Denmark/190/2008	17/04/08	MDCK2 \1	40	80	80	<	160	160	80	320	160	160
A/Ulan-Ude/2/2008	21/04/08	C2 \1	80	80	80	<	80	160	40	80	80	80
A/Lyon/CHU/ 19.113/08	05/05/08	MDCK2 \1	<	<	40	<	80	160	80	160	80	40
A/Iceland/82/2008	06/05/08	MDCK2 \1	40	40	80	<	80	160	40	160	160	40
A/Johannesburg/15/2008	26/06/08	MDCKx \1	160	80	320	80	160	320	320	640	160	320

Vaccine strain

¹ < = <40

Table 4B. Antigenic analyses of influenza A H3N2 viruses

Haemagglutination inhibition titre ¹														
Viruses	Collection Date	Passage History	Post infection ferret sera											
			A/Wis 67/05 F6/06	A/Nepal 921/06 F2/07	A/Trieste 25c/07 F5/07	A/Prag 3/07 F9/07	A/Tri 25E/07 F11/07	A/Wis 3/07 F15/07	A/Hen 147/07 F16/07	A/Bris 10/07 F18/07	A/Uru 716/07 F3/07	A/Fin 9/08 F12/08	A/HK 1050/08 F23/08	A/Jbg 15/08 F22/08
REFERENCE VIRUSES														
A/Wisconsin/67/2005	31/8/05	SpfCk3, E3 V4	5120	5120	2560	640	2560	1280	5120	2560	5120	2560	2560	2560
A/Nepal/921/2006	11/7/06	E2 V2	1280	5120	2560	1280	2560	640	5120	1280	2560	2560	2560	5120
A/Trieste/25c/2007	Jan-07	MDCKx V4	<	80	80	<	160	160	80	80	80	80	80	80
A/Prague/3/2007	Feb-07	MDCK3, E4 V2	1280	5120	2560	1280	2560	640	5120	640	2560	2560	1280	2560
A/Trieste/25E/2007	Jan-07	MDCKx V2,E2	1280	2560	1280	320	1280	320	5120	640	2560	1280	1280	640
A/Wisconsin/3/2007	21/1/07	Ex V6	5120	5120	5120	1280	2560	2560	5120	2560	5120	2560	5120	2560
A/Henan/Jinshui/147/2007	unknown	E3 V1	1280	2560	1280	640	1280	320	5120	640	2560	1280	1280	640
A/Brisbane/10/2007	6/2/07	E2 V1	1280	5120	2560	1280	2560	1280	5120	1280	2560	2560	2560	1280
A/Uruguay/716/2007	21/6/07	SpfCk1, E3 V2	2560	5120	2560	640	2560	1280	5120	1280	5120	2560	2560	1280
A/Finland/9/2008	7/1/08	MDCK2 V3	<	80	160	<	160	160	160	160	80	320	160	160
A/Hong Kong/1050/08*	23/4/08	MDCK2 V1	<	160	160	<	160	160	80	80	80	80	160	80
A/Johannesburg/15/2008*	26/6/08	MDCKx V2	<	80	160	<	160	160	160	80	80	160	160	160
TEST VIRUSES														
A/Hong Kong/796/08*	15/03/08	MDCK2 V1	320	160	160	40	320	320	640	640	160	320	ND	ND
A/Geneva/8428/09*	22/01/08	MDCK3 V1	40	80	40	<	80	80	40	160	80	ND	ND	ND
A/Lyon/CHU19.113/08*	05/05/08	P2MDCK V1	<	<	40	<	80	160	80	160	80	40	ND	ND
A/Mauritius/402/2008*	09/07/08	MDCK3	<	80	80	<	80	80	40	40	40	40	80	40
A/Geneva/8417/2008	16/01/08	MDCK1 V1	<	80	80	<	80	80	40	40	40	40	80	40
A/Lyon/762/08	26/02/08	P2MDCK V1	40	80	160	40	160	160	160	80	40	80	160	80
A/Toulouse/609/08	18/01/08	P2MDCK V1	40	80	160	40	160	160	160	80	40	80	160	80
A/Denmark/190/2008	17/04/08	MDCK2 V1	<	80	160	<	160	160	80	80	40	160	160	80
A/Finland/149/2008	28/01/08	MDCK2 V1	<	80	160	<	160	160	80	80	40	160	160	80
A/Denmark/183/2008	11/04/08	MDCK2 V1	<	40	40	<	80	80	40	40	40	40	80	40
A/Denmark/185/2008	16/04/08	MDCK2 V1	<	40	40	<	80	80	40	40	40	40	80	40
A/Finland/122/2008	25/01/08	MDCK1 V1	<	40	80	<	80	160	80	80	40	160	160	80
A/Norway/848/2008	01/04/08	MDCK2 V1	<	80	80	<	80	80	ND	80	80	80	ND	80
A/Norway/966/2008	15/04/08	MDCK2 V1	<	40	40	<	40	80	ND	<	40	40	ND	40
A/Norway/1034/2008	19/04/08	LLC-MK2-1 V1	40	80	80	<	80	160	ND	80	160	80	ND	80
A/Hong Kong/1778/2008	23/07/08	MDCK2 V1	160	160	320	80	160	320	ND	160	320	320	ND	320
A/Hong Kong/1859/2008	25/07/08	MDCK2 V1	40	80	80	<	160	160	ND	80	160	80	ND	80
A/Ghana/11/2008	15/07/08	MDCKx V1	40	80	80	<	160	320	ND	160	160	160	ND	160
A/Washington/42/2007	12/12/07	E6/E2 V1E	320	80	320	40	80	320	ND	640	320	320	ND	320
A/Mississippi/4/2008	15/01/08	E5/E2 V1E	640	320	1280	160	640	640	ND	1280	1280	640	ND	1280
A/Perth/201/08	15/07/08	E4 V1E	640	1280	1280	320	640	320	ND	320	1280	640	ND	320
A/Brisbane/24/08	23/06/08	E5 V1E	2560	2560	2560	640	2560	2560	ND	1280	5120	2560	ND	2560
A/Wellington/2/08	28/05/08	E3 V1E	2560	1280	2560	640	1280	2560	ND	2560	2560	1280	ND	2560
A/Singapore/21/08	02/05/08	E3 V1E	1280	2560	1280	320	1280	640	ND	640	2560	1280	ND	640
A/Ulaanbaatar/1600/2008	13/02/08	C2/C2 V1	40	80	160	<	160	80	ND	160	80	80	ND	160
A/Toyama/123/2008	03/04/08	MDCK1+2 V1	40	80	160	40	160	320	ND	160	160	320	ND	160
A/Saitama/27/2008	08/05/08	MDCK1+2 V1	160	160	160	<	160	160	ND	160	160	160	ND	80

Vaccine strain

¹ < = <40; * = Viruses subjected to neutralisation assays (see Table 5); ND = Not done.

Table 5. Antigenic analysis of influenza AH3N2 viruses by virus neutralisation

Viruses	Collection Date	Passage History	Virus Neutralisation titre ¹									
			Post infection ferret sera									
			A/Panama	A/Fujian	A/Well	A/Wisc	A/Nepal	A/Trieste	A/Henan	A/Bris	A/Urug	A/Fin
			2007/99 F17/01	411/02 F6/03	1/04 F13/04	67/05 F6/06	921/06 F2/07	25c/07 F5/07	147/07 F16/07	10/07 F18/07	716/07 F3/07	9/08 F12/08
REFERENCE VIRUSES												
A/Panama/2007/99	unknown	E	2560	160	160	<	<	<	<	<	ND	ND
A/Fujian/411/02	unknown	MDCK	160	640	80	160	160	80	160	160	80	160
A/Wellington/1/04	26/01/04	E3 1	320	160	2560	320	320	160	320	320	320	320
A/Wisconsin/67/05	31/08/05	SpfCk3, E3 14	40	160	640	2560	2560	2560	2560	2560	ND	ND
A/Nepal/921/06	11/07/06	E2 12	<	160	320	1280	5120	1280	5120	2560	ND	ND
A/Trieste/25c/07	Jan-07	MDCKx 14	40	160	320	1280	2560	2560	2560	2560	ND	ND
A/Henan/147/07	unknown	E3 1	<	160	320	1280	1280	640	2560	1280	ND	ND
A/Brisbane/10/07	06/02/07	E2 1	<	80	640	2560	2560	2560	2560	2560	ND	ND
A/Uruguay/716/07	21/06/07	SpfCk1, E3 12	<	40	80	320	320	320	1280	320	640	320
A/Finland/9/08	07/01/08	MDCK2 13	40	80	160	640	1280	1280	2560	1280	2560	2560
TEST VIRUSES												
A/Hessen/6C/07	09/02/07	MDCKx 13	40	160	640	1280	5120	2560	5120	2560	2560	ND
A/Lyon/1381/07	18/12/07	MDCK2 1	40	80	80	320	160	640	320	640	640	ND
A/Berlin/20C/07	09/02/07	MDCKx 12	<	80	160	640	1280	1280	1280	1280	1280	640
A/Geneva/8428/08	22/01/08	MDCK3 1	0	80	80	320	1280	640	1280	640	640	ND
A/Hong Kong/796/08	15/03/08	MDCK2 1	80	160	320	1280	2560	2560	2560	2560	2560	2560
A/Hong Kong/1050/08	23/04/08	MDCK2 1	80	160	320	1280	2560	2560	2560	1280	2560	1280
A/Lyon/CHU19.113/08	05/05/08	P2MDCK 1	40	160	320	1280	2560	2560	2560	2560	2560	2560
A/Johannesburg/15/08	26/06/08	MDCKx 1	<	160	160	640	2560	2560	2560	1280	2560	1280
A/Mauritius/402/08	09/07/08	MDCK3 1	<	80	160	1280	1280	2560	1280	1280	2560	1280

Vaccine strain

¹ Based on 50% plaque reduction compared to serum negative controls; ND = Not done.

Figure 3. Phylogenetic comparison of H3 HA genes

Vaccine strains

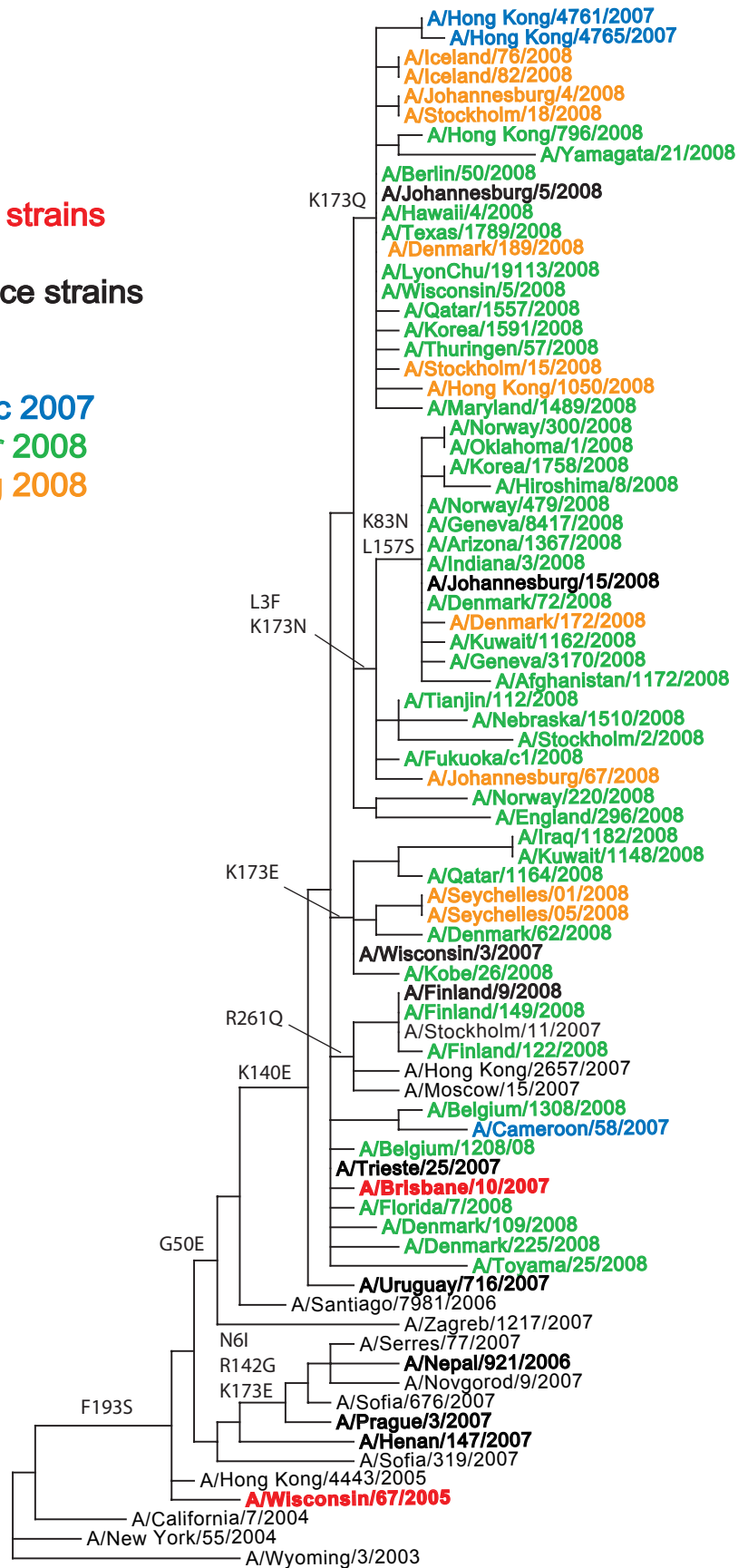
Reference strains

Isolates

Aug-Dec 2007

Jan-Mar 2008

Apr-Aug 2008



10 nucleotide substitutions

Figure 4. Phylogenetic comparison of N2 neuraminidase genes

Vaccine strain

Reference strains

Isolates

Aug-Dec 2007

Jan-Mar 2008

Apr-Aug 2008

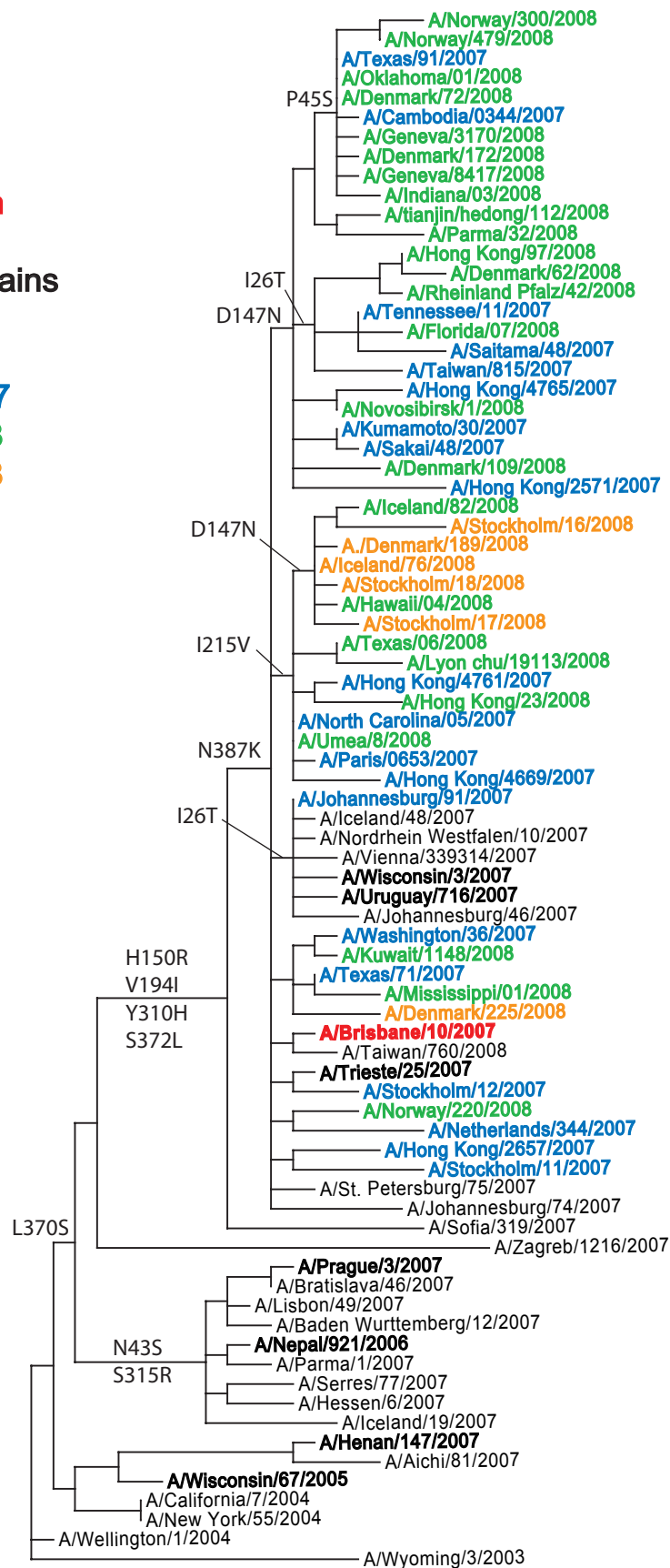


Table 6A. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Collection date	Passage	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			B/Jiang 10/03 F8/04	B/Eg 144/05 F3/07	B/FI 4/06 F20/07	B/Bris 3/07 F24/07	B/Sen 114/07 F1/08	B/Barc* 143/08 F11/08	B/Eng* 145/08 F9/08	B/Vall* 18/08 F10/08
REFERENCE VIRUSES										
B/Jiangsu/10/2003	12/9/2003	E4 1/3	640	80	40	80	40	<	40	10
B/Egypt/144/2005	5/1/2005	E2 1/2	20	160	320	640	160	20	40	10
B/Florida/4/2006	15/12/2006	E2 1/1	20	160	320	640	320	20	40	10
B/Brisbane/3/2007	9/3/2007	E2 1/2	40	160	640	640	320	20	80	20
B/Sendai-H/114/2007		E1+2 1/1	20	80	160	320	320	10	80	10
B/Barcelona/143/2008	06/01/08	MDCK1 1/2	40	80	80	160	80	160	160	160
B/England/145/2008		Ex 1/2	<	<	<	40	40	10	80	40
B/Valladolid/18/2008	18/02/08	MDCK1 1/2	40	80	40	80	80	80	80	160
TEST VIRUSES										
B/Paris/2400/2008	Mar-08	MDCK1 1/1	160	40	320	160	160	160	160	160
B/Rzeszow-PL/11/2008	12/03/08	MDCK 1/1	320	160	160	320	160	320	320	320
B/Berlin/33/2008	12/03/08	MDCKx 1/1	160	160	320	320	320	160	320	320
B/Ireland/3421/08v	25/03/08	MDCKx 1/1	160	80	160	160	80	160	160	320
B/Latvia/4-246/2008	06/04/08	MDCKx 1/1	20	80	40	40	40	40	80	160
B/Iceland/66/2008	13/04/08	MDCK1 1/1	80	160	160	160	160	160	160	320
B/Bratislava/411/2008	15/04/08	MDCK1 1/1	80	160	320	320	320	80	160	160
B/St Petersburg/94/2008	21/04/08	C1 1/1	320	160	80	160	160	320	160	320
B/Finland/246/2008	21/04/08	MDCK1 1/1	160	160	160	320	320	80	160	160
B/Denmark/95/2008	28/04/08	MDCK2 1/1	40	80	40	40	80	80	160	160
B/Lyon/999/08	29/04/08	MDCK3 1/1	80	80	80	80	160	40	80	80
B/Hong Kong/1119/2008	13/05/08	MDCK2 1/1	160	160	160	320	320	160	80	320
B/Johannesburg/18/2008	29/05/08	MDCKx 1/1	80	160	160	320	320	80	320	160
B/Cape Town/30/2008	30/05/08	MDCKx 1/1	320	160	320	320	320	320	320	320
B/La Plata/53583/2008	18/06/08	M1 1/1	80	160	80	160	160	80	80	160
B/Buenos Aires./819667/2008	18/06/08	M1 1/1	80	80	80	80	160	40	80	80

Vaccine strains

¹ < = <40; * < = <10

Table 6B. Antigenic analyses of influenza B viruses (Yamagata lineage)

Haemagglutination inhibition titre ¹											
Viruses	Collection date	Passage	Post infection ferret sera								
			B/FI	B/Eg	B/FI	B/Bris	B/Sen	B/Barc*	B/Eng*	B/Vall*	B/Bang
			04/06 SH478	144/05 F3/07	4/06 F19/07	3/07 F24/07	114/07 F1/08	143/08 F11/08	145/08 F9/08	18/08 F10/08	3333/07 F21/08
REFERENCE VIRUSES											
B/Egypt/144/2005	5/1/2005	E2 √2	1280	160	320	640	320	10	40	10	160
B/Florida/4/2006	15/12/2006	E2 √1	2560	160	320	640	640	<	80	10	160
B/Brisbane/3/2007	9/3/2007	E2 √2	2560	160	320	640	320	10	80	10	160
B/Sendai-H/114/2007		E1+2 √1	640	40	40	160	160	<	40	<	<
B/Barcelona/143/2008	06/01/08	MDCK1 √2	5120	160	640	320	320	320	320	320	320
B/England/145/2008		Ex √2	160	<	<	<	40	<	80	<	<
B/Valladolid/18/2008	18/02/08	MDCK1 √2	2560	80	80	80	80	80	160	320	160
B/Bangladesh/3333/2007	19/8/07	E4 √1	640	40	40	80	40	10	20	10	160
TEST VIRUSES											
B/Norway/866/2008	10/04/08	MDCK1 √1	5120	80	160	160	160	160	320	320	320
B/Norway/878/2008	02/04/08	MDCK1 √1	5120	80	320	320	320	160	160	160	160
B/Norway/1035/2008	25/04/08	MDCK1 √1	5120	80	320	320	320	160	160	320	320
B/Norway/1062/2008	10/04/08	MDCK1 √1	2560	80	<	80	160	20	40	80	40
B/Norway/1095/2008	09/05/08	MDCK1 √1	2560	80	40	80	80	160	160	160	160
B/Hong Kong/1208/2008	30/06/08	MDCK2 √1	2560	80	160	160	160	160	160	160	160
B/Hong Kong/1264/2008	23/07/08	MDCK2 √1	1280	80	320	320	160	160	160	160	320
B/Cape Town/111/2008		MDCK √1	2560	40	80	160	160	80	40	80	80
B/Johannesburg/321/2008		MDCK √1	5120	160	1280	320	320	320	320	320	320
B/Johannesburg/342/2008		MDCK √1	5120	160	320	320	320	160	160	320	320
B/Brisbane/1/2008	15/01/08	E3 √1E	1280	80	160	640	320	<	40	10	80
B/Brisbane/9/2008	31/03/08	E6 √1E	640	40	40	160	80	10	40	10	80
B/Perth/6/2008	09/04/08	E5 √1E	1280	80	160	640	320	<	40	<	80
B/Perth/204/2008	11/07/08	E3 √1E	640	40	40	160	160	<	80	<	<
B/Ulaanbaatar/1798/2008	15/02/08	E3+3 √1E	320	<	<	40	40	<	40	<	<
B/Indiana/01/2008	10/03/08	E2 √1	640	<	<	80	<	<	20	10	80
B/Santiago/5241/2008	12/04/08	E4/E2 √1E	640	40	<	<	40	10	40	20	40
B/Tottori/3/2008	08/03/08	MDCK1+2 √1	1280	80	160	160	80	160	80	160	160
Vaccine strains											

¹ < = <40; * < = <10; ND = Not done

Figure 5. Phylogenetic comparison of BHA genes (Yamagata lineage)

Vaccine strains

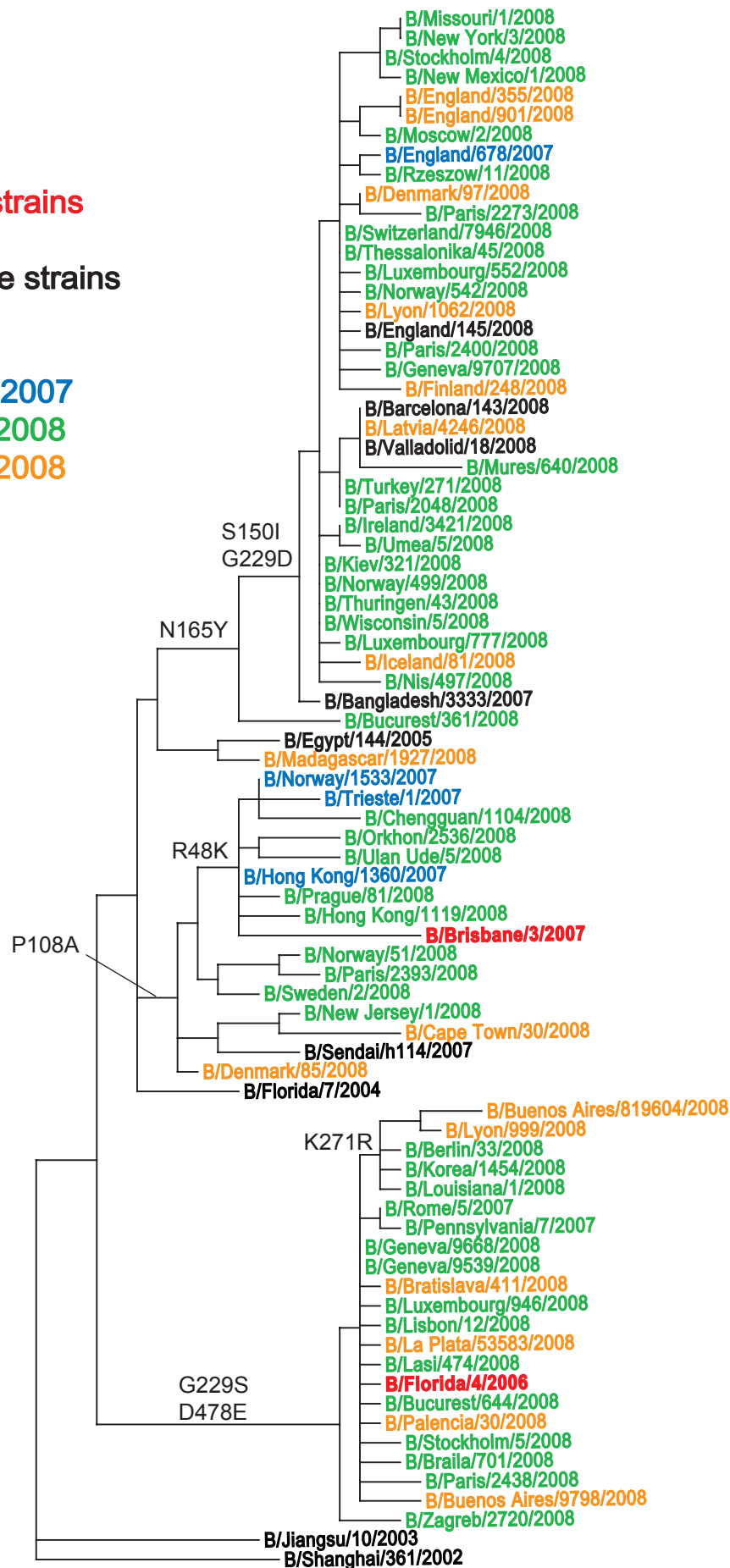
Reference strains

Isolates

Aug-Dec 2007

Jan-Mar 2008

Apr-Aug 2008



10 nucleotide substitutions

Figure 6. Phylogenetic comparison of B NA genes

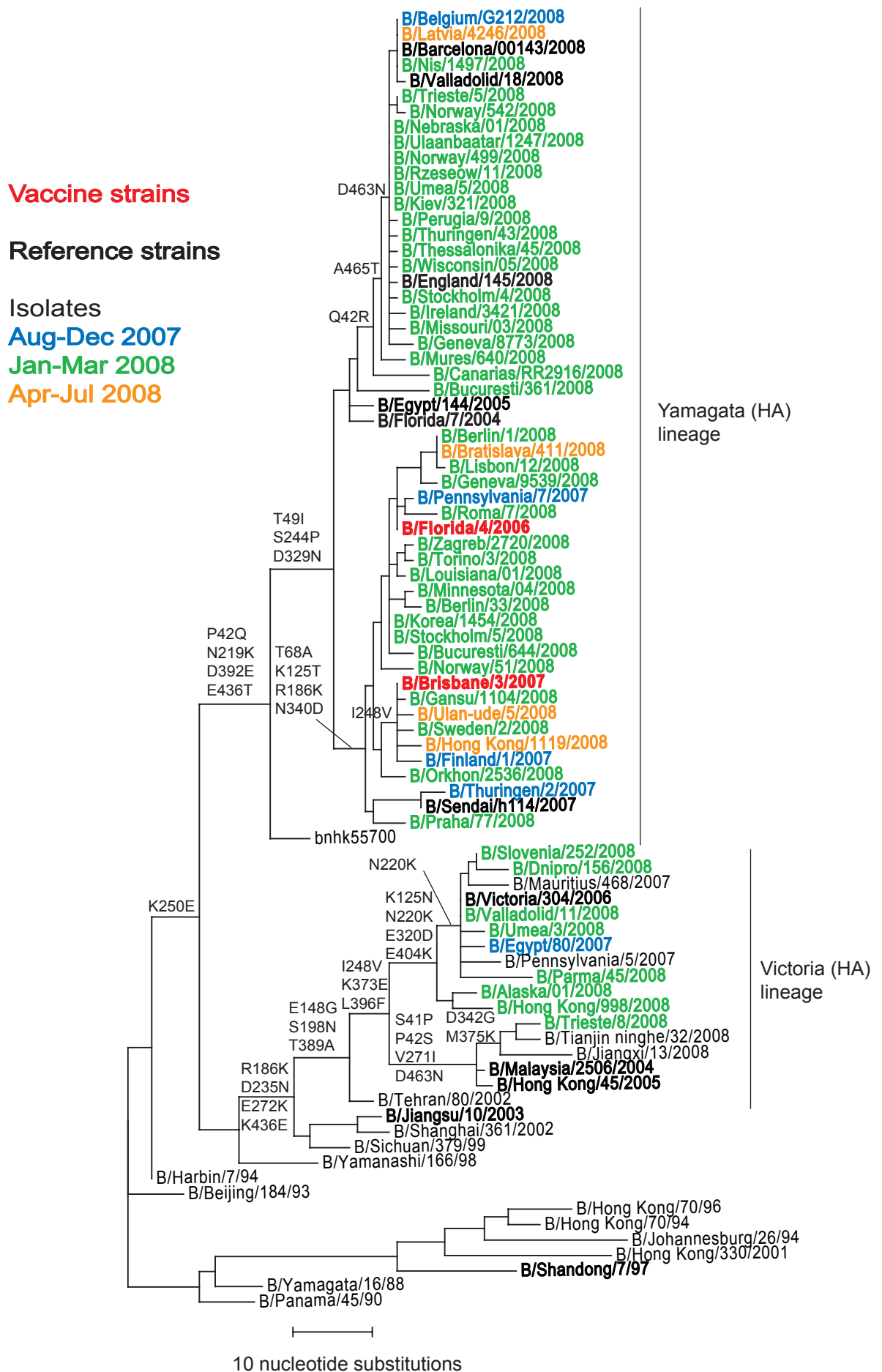


Table 7. Antigenic analyses of influenza B viruses (Victoria lineage)

Haemagglutination inhibition titre ¹								
Viruses	Isolation Date	Passage History	Post infection ferret sera					
			B/Shan	B/Shan	B/Bris*	B/HK	B/Mal	B/Vic
			7/97 SH	7/97 F12/02	32/02 F20/03	45/05 F12/05	2506/04 F28/05	304/06 F16/06
REFERENCE VRUSES								
B/Shandong/7/97		Ex	2560	320	80	160	160	160
B/Brisbane/32/2002	02/07/02	E2 \6	1280	80	80	80	160	80
B/HK/45/2005	07/02/05	MDCK2 \4	2560	320	160	160	320	160
B/Malaysia/2506/2004	06/12/04	E3 \3	1280	160	80	80	320	160
B/Victoria/304/2006	06/06/06	E2 \2	1280	160	160	160	160	160
TEST VIRUSES								
B/Denmark/1/2008	04/01/08	MDCK2 \1	5120	40	40	320	80	160
B/Slovenia/252/2008	17/01/08	MDCK1 \1	1280		<	160	<	<
B/Valladolid/11/2008	23/01/08	MDCK1 \2	1280	80	<	320	40	40
B/Dnipro/156/2008	11/02/08	MDCK1 \1	1280	80	<	320	<	40
B/Parma/45/08	13/02/08	MDCKx \1	1280	160	<	320	40	40
B/Geneva/9997/2008	10/03/08	MDCK1 \1	5120	320	160	160	320	160
B/ Kiev/45/08	14/03/08	MDCK3 \1	2560	80	<	320	<	<
B/Baden-Wuerttemberg/106/2008	17/03/08	MDCKx \1	640	40	<	40	<	<
B/Morocco/404H/2008	03/04/08	MDCKx \1	1280	<	<	160	<	<
B/Hong Kong/998/2008	05/04/08	MDCK2 \1	2560	40	<	160	<	<
B/Johannesburg/8/2008	14/04/08	MDCKx \1	5120	40	<	160	40	<
B/Johannesburg/43/2008	10/06/08	MDCKx \1	2560	80	<	160	<	<
B/Toulouse/620/2008	29/01/08	MDCK2 \1	1280	<	<	80	<	<
B/Poitiers/985/2008	19/03/08	MDCK2 \1	2560	<	<	160	<	<
B/Jiangxi Donghu/1792/2007	26/12/07	E2/E2 \1E	640	80	80	80	160	80
B/Hubeijiangnan/1111/2008	07/04/08	E2+2 \1E	640	80	80	80	160	80
B/Hubeijiangnan/1146/2008	28/04/08	C1+2 \1	1280	<	<	160	<	<

¹ < = <40; SH = hyperimmune sheep serum; * < = <10

Figure 7. Phylogenetic comparison of B HA genes (Victoria lineage)

Vaccine strains

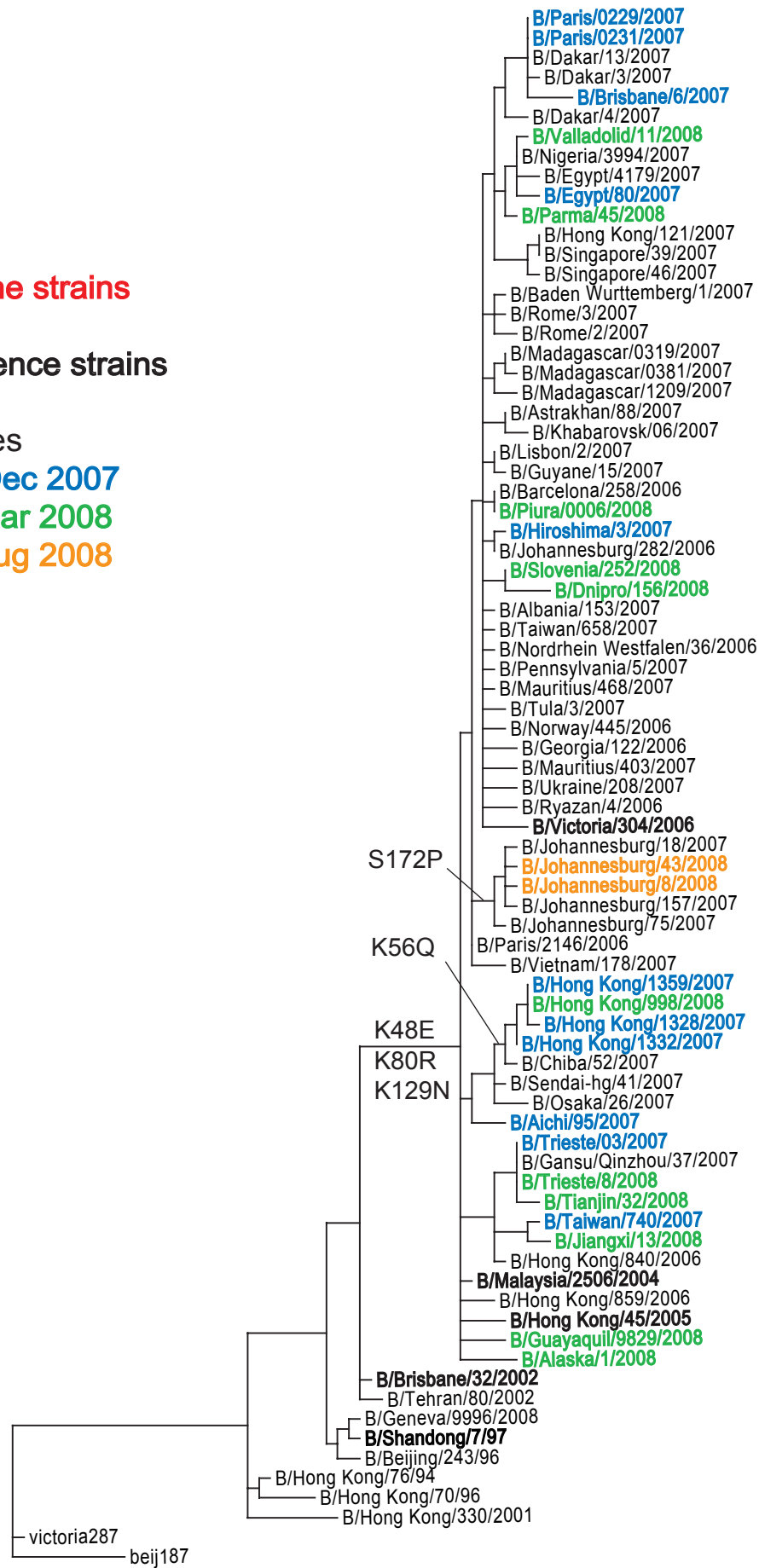
Reference strains

Isolates

Aug-Dec 2007

Jan-Mar 2008

Apr-Aug 2008



10 nucleotide substitutions

Table 8. Amino acid changes characteristic of recent B sequence variants

HA Lineage	Representative strain	Amino acid changes	
		HA	NA
Yamagata ¹	B/England/145/08	N165Y S150I G229D	Q42R D463N A465T
	B/Brisbane/3/07	P108A (R48K)	T68A K125T R186K N340D
	B/Florida/4/06	G229S D478E (K271R)	T68A K125T R186K N340D
Victoria ²	B/Victoria/304/06	(K56Q) (S172P)	P41S S42P K125N N220K I271V E320D E404K N463D

¹ Relative to B/Florida/7/04; ² Relative to B/Malaysia/2506/04.